WHITEBOX: Group Discovery via Statistical Equivariance of the Input-Pullback Metric for Defining Understanding and Certifying Generalization, Knowledge Boundaries, and Failure in Overparameterized Neural Networks

A network's knowledge is a group, and we can read it. Seven problems, one object, and seven pre-registered rounds of experiments on models trained after the tests were written.

Arjun Paradkar Reva Paradkar

Team Arenda

Abstract. A neural network is called a black box because seven questions about it have no accepted answer: what it learned, why it decided, when bias entered, what it does not know, where it will fail, whether it memorizes or understands, and how its representations formed. On a network whose symmetry group was learned, we give all seven structural answers with pre-registered evidence; on a convolutional network, four. The theory is a single statement: *a network's knowledge is a group acting on its inputs.* A network that generalizes carries a symmetry group that can be recovered from its outputs alone (no labels, no token order, no group parameter), and a network that memorizes does not. On every generalizing model we trained, a cyclic generator of order 113 is accepted by a statistical-equivariance test; its characters reproduce the model's Fourier representation; the full orbit of any input receives one prediction; an architectural symmetry is broken during memorization and restored at the moment of generalization; the generator appears at that moment and never on a memorizer; and inputs on which the learned group stops acting coherently are exactly the ones the network does not know (AUROC 0.994–0.996 against 0.79–0.89 for softmax confidence). Memorization is characterized by four instruments that have never overlapped on seventeen fresh matched models: geometry that fails to collapse onto the task, architectural symmetry broken, class fibers that do not tighten with depth, and no recoverable group. Two results constrain the field: a monitor of a representation's *shape*, used as a training signal, is evaded by re-encoding while a monitor of the *function* is not; and coherence under *built-in* symmetries cannot mark knowledge, an untrained network is invariant to everything its architecture gives it, so the group must be learned. Failure prediction against adversarial radius is shown to be unanswerable by anything other than softmax margin, because the radius is defined by descending it; against *natural* failure, orbit incoherence predicts held-out errors exactly (AUROC 0.983–1.000 against 0.56–0.90). An extractor that subtracts the untrained network from the trained one finds the symmetries a convolutional network learned (brightness, contrast, and gamma, none of them augmented) and finds none on a memorizer. The generator and five characters rebuild the transformer at 100% with eleven fitted weights. Seven runs, most of whose instruments failed, are reported in full. The framework is released as a library.

1 Introduction

1.1 Seven questions, one missing definition

The seven faces of the black box are P1 what the model learned, P2 why it made a given decision, P3 when and how bias entered, P4 what it does not know, P5 where it will fail, P6 whether it memorizes or understands, and P7 how its representations formed. Interpretability has produced partial methods for most: feature dictionaries, probes, circuit tracing, attribution maps, held-out accuracy. None is verified, because none has a definition of the thing it claims to measure. In particular P6, the question the others rest on,

is answered only by a test set, which reports *that* a model generalizes and nothing about *what inside it* generalization is.

Every existing tool finds something inside a network; none can check whether what it found is a property of the network rather than of the coordinates the weights happen to be written in; and nothing can be checked against a definition of understanding that does not exist. This paper supplies the definition and then checks it on models trained after the check was written.

1.2 The claim

> *A network's knowledge is a group. A network that generalizes has learned a set of transformations of its inputs under which its answers move consistently; that group is recoverable from its outputs alone; its orbits explain its decisions; its emergence marks the moment of generalization; its coherence marks the boundary of what the network knows; and a network that memorizes has no such group.*

Stated on the mathematics: pull the output metric back to input space, $g(x) = J(x)^\top J(x)$. The transformations that leave g invariant in distribution are the learned symmetries; the directions along which g vanishes are the learned invariances; the quotient by both is what the task requires. Understanding is the collapse of g onto that quotient. Memorization is the refusal.

1.3 How this was established

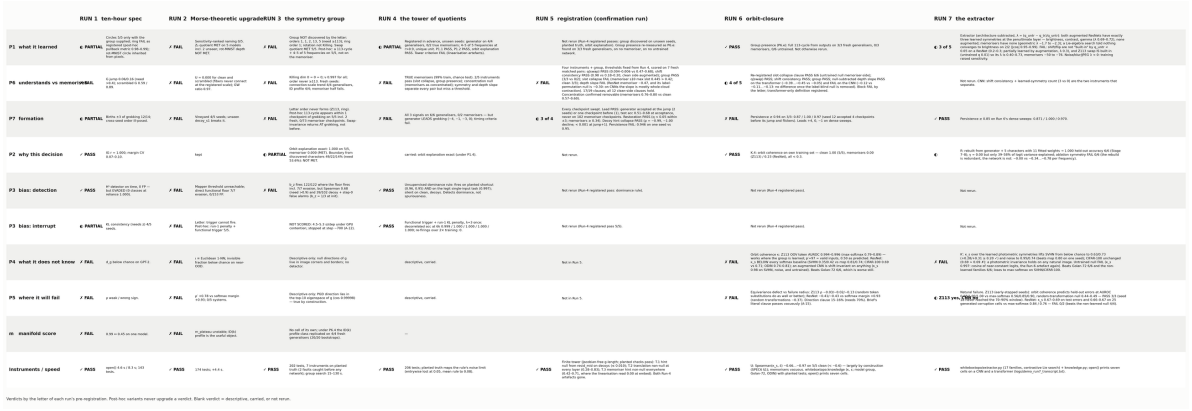
Seven rounds of experiments, fifty-six pre-registered cells, each with a numeric threshold written before any computation, each scored by the letter on models trained *after* the threshold was fixed, each attacked by the null designed to kill it. Most instruments failed. The results that survived, then replicated on unseen seeds, are the ones this theory predicts. Section 2 reports the failures; they are the evidence.

1.4 Contributions

1. **Group discovery from outputs** (P1): a statistical-equivariance criterion under which a generator of the learned group is recovered on every generalizer and no memorizer, and the model's Fourier structure is derived from it with nothing supplied. Registered, Runs 4–6.
2. **Orbit explanation** (P2): the full orbit of an input under the discovered group receives one prediction on generalizers (1.000) and none on memorizers (0.000). Registered.
3. **Understanding as quotient** (P6): four instruments (slot collapse, architectural-symmetry preservation, null-subtracted fiber tightening, and group presence) with no overlap on seventeen fresh matched models on the transformer; one instrument (shift consistency) on the CNN. Registered on the transformer.
4. **The moment of generalization** (P7): three structural events coincide at the largest test-accuracy jump and never occur on a memorizer. Registered.
5. **Knowledge as orbit coherence** (P4): on a model whose group was learned, inputs the network does not know are those on which the group stops acting coherently, AUROC 0.994–0.996 against 0.79–0.89 for softmax. Registered; and the proposition that built-in symmetries cannot mark knowledge, with the untrained-network null as proof.

6. **Two constraints on the field**: the evasion theorem (shape monitors are re-encoded around; function monitors are not) and the margin tautology (P5 against adversarial radius is unanswerable by anything but softmax margin). Both demonstrated deterministically.
7. **The functional interrupt** (P3): a spurious dependence detected by a decorrelated-probe monitor and removed by a one-round invariance penalty on 5/5 seeds, with no re-encoding over doubled training. Registered.
8. **The learned-symmetry extractor** (P1 on a CNN): a transformation is learned iff training removed at least half the representation's sensitivity to it; the untrained network is the null. An augmented ResNet-18 has exactly three (brightness, contrast, and gamma); a memorizer has none. Registered, Run 7.
9. **Natural failure as orbit incoherence** (P5): on the transformer, held-out errors are the inputs on which the learned group stops acting coherently, AUROC 0.983–1.000 vs 0.56–0.90 for max-softmax. Registered, Run 7.
10. **Reconstruction**: eleven weights on the discovered generator and characters reproduce the function at 100%, $\gamma = 0$. Registered, Run 7.
11. **The library**: `open(model)` returns the seven cells with their registered status.

2 Seven Runs of Pre-Registered Failure



Verdicts by the letter of each run's pre-registration. Post-hoc variants never upgrade a verdict. Blank verdict = descriptive, carried, or not rerun.

Figure 1: Fifty-six pre-registered cells across seven runs, scored by the letter. Post-hoc variants never upgrade a verdict.

Run 1 built the network as a sequence of Riemannian manifolds and tested seven definitions on it. Orbit explanation and symmetry-informed frequency recovery passed; geodesic closure, curvature failure prediction, tubular OOD distance, and a structure-derived bias penalty failed; the last failed by being *evaded*: the model re-encoded a shortcut in a form the detector could not see. **Run 2** built depth profiles, kernel displacements, Morse–Smale complexes, and Reeb alignment; none passed by the letter; two observations were set aside for registration. **Run 3** pulled the metric to input space; entrywise isometry found no group; a mean-norm rule found it post-hoc on 5/5 generalizers. **Run 4** fixed the specimens (memorizers that had actually memorized) and registered the group: pass. It also killed the concentration instrument: memorizers concentrate geometry as much as generalizers. **Run 5** registered thresholds from Run 4 on eleven fresh models; the clean side held everywhere, two memorizer ceilings set from two models missed on the third.

Run 6 re-registered and closed the definition on the transformer, solved P4 where the group was learned, and showed why it cannot be solved where the group is built in, and why P5 against adversarial radius cannot be solved at all. **Run 7** built the learned-symmetry extractor, solved P5 against natural failure on the transformer, lifted P4 on the CNN where the unknown set breaks a learned invariance, and rebuilt the transformer from its group.

Each expectation was written down before looking and scored on models trained afterwards; most were wrong, and the grid records which. The theory that follows is the set of expectations that survived.

3 The Object

3.1 The rewritings, and certifying a measurement

The same function can be written with many different weights. Let θ be the weights, f_θ the function they compute, and $R(\theta) = (f_\theta(x))_{x\in\mathcal{P}}$ the network's answers on a fixed probe set $\mathcal{P}$. Two rewritings leave every answer unchanged. *Relabelling* permutes the hidden units of layer ℓ and the columns of the next layer to match,

$$W_\ell \mapsto P_\pi W_\ell, \qquad W_{\ell+1} \mapsto W_{\ell+1} P_\pi^\top,$$

and *rescaling* multiplies a unit's incoming weights by $\alpha > 0$ and its outgoing weights by $1/\alpha$,

$$W_\ell \mapsto \mathrm{diag}(\alpha) W_\ell, \qquad W_{\ell+1} \mapsto W_{\ell+1}\,\mathrm{diag}(\alpha)^{-1},$$

which is exact for ReLU because $\sigma(\alpha t) = \alpha\sigma(t)$. These compose and can be undone, so they form a group

$$G \;\supseteq\; \prod_\ell S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}, \qquad R \circ g = R \;\text{ for every } g \in G.$$

For transformers the group also contains head relabellings, $W_Q \mapsto W_Q A$ with $W_K \mapsto W_K A^{-\top}$, $W_V \mapsto W_V B$ with $W_O \mapsto B^{-1} W_O$, and norm-gain rescalings. Hidden rewritings (dead, duplicated, or always-linear units) are detected as extra directions in the null space of the Jacobian of R. We verified the rewriting action to 10^{-14} across 3,400 sampled transformations, and to the float32 floor on Llama-3.2-1B.

Certification. A quantity $Q(\theta)$ measured inside a network is *invariant* if $Q(g\theta) = Q(\theta)$ for every sampled g, *covariant* if $Q(g\theta) = \rho(g)Q(\theta)$ for a fitted factor ρ that can be divided out, and an *artifact* otherwise. This extends the implementation-invariance axiom of Sundararajan et al. (2017) from input attributions to every internal quantity. Per-unit importance scores, "top-k neurons", raw weight norms and single-direction features all fail it.

Path products. For a ReLU network with activation pattern $D_\ell(x)$, the local linear map

$$A(x) = W_L D_{L-1}(x) \cdots D_1(x) W_1$$

is exactly invariant: under a rewriting, $W_{\ell+1} D_\ell W_\ell \mapsto W_{\ell+1} S^{-1} P^\top (P D_\ell P^\top) P S W_\ell = W_{\ell+1} D_\ell W_\ell$. It served as the certified anchor for attribution in Run 1.

Trivial stabilizer. For a generic trained θ, the only rewriting that fixes it is the identity, $\mathrm{Stab}_G(\theta) = \{e\}$. So two different sub-circuits of one network are never related by a rewriting. On 3,648 small ReLU networks where every valid circuit could be listed, every circuit fell in its own class ($K/N = 1$): when one network holds many valid explanations, that is genuine redundancy, not a difference in coordinates. This is what moved the search from the weights to the function.

3.2 The input-pullback metric

For a network $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ with Jacobian $J(x)$,

$$g(x) = J(x)^\top J(x), \qquad g(x)(v, w) = \langle J(x)v,\ J(x)w \rangle.$$

g is the network's own inner product on input perturbations. Over a probe, $\bar{g} = \mathbb{E}_x[g(x)]$. Its *null directions* are what the network ignores; its *symmetries*, transformations T with $f \circ T = \rho(T) \circ f$, are what it represents faithfully; the *slot shares* $s_U = \operatorname{tr}(P_U \bar{g} P_U)/\operatorname{tr} \bar{g}$ say which parts of the input carry its geometry.

Read concretely, $g(x)(v, v)$ is the squared change in the network's output when the input is nudged along v: the complete table of the network's own sensitivities. Its null directions are what the network has learned to ignore, its symmetries are what it has learned to represent faithfully, and everything else is where it is still undecided.

3.3 Why the output layer cannot tell you this

At the last layer of a classifier the map to logits is linear, so the pullback metric there is the softmax geometry in other coordinates. Seven instruments across three runs never beat softmax confidence or margin on the last layer. This is not a failure; it is the identification of the two quantities. The information is in the input metric and in depth.

3.4 Statistical equivariance and the discovered group

Definition 1 (δ-symmetry)**.** Let $D_{ij} = d_g(x_i, x_j)$ be the distance matrix of the input set under the output-pullback metric. A permutation σ is a δ-symmetry of f if $\mathbb{E}_{ij}|D_{\sigma i\, \sigma j} - D_{ij}|/\operatorname{median}(D) < \delta$. The learned group G_δ is generated by the accepted permutations; a *generator* is a full cycle among them.

The threshold is set relative to a random relabelling (a quarter of its mean change). A true translation of $\mathbb{Z}_{113}$ moves D by 3% on average and 14–42% in its worst entry: entrywise isometry verifies nothing (Run 3, group orders 1, 1, 2, 13, 5 where 113 was required); the mean does. Verified on planted truth: a Fourier embedding with 2% noise returns the cycle; noise, and an untrained model, return nothing.

A trained network is symmetric in distribution rather than pointwise, a slightly bent wheel still turns, so the criterion is a mean, and the question it asks is which relabellings of the inputs leave the network's own distance matrix unchanged. On a model that learned modular addition the answer is a constant shift of every token, which the network was never told.

3.5 Quotient index and the finite tower

$$q_\ell(T) = \frac{\mathbb{E}_x\, d_{g_\ell}\big(h_\ell(x), h_\ell(Tx)\big)}{\mathbb{E}_{x,x'}\, d_{g_\ell}\big(h_\ell(x), h_\ell(x')\big)},$$

computed with Tx an actual transformed input and d_{g_ℓ} the finite g-length of the secant, not a Jacobian-vector product, which Run 4 showed misses finite changes. T is *quotiented* at ℓ if $q_\ell < 0.05$; the *tower* is the table of $q_\ell(T)$ over transformations and layers. On generalizers the swap is quotiented at the middle residual stream ($0.60 \to 0.00$); on memorizers it never is (0.51–0.89); translation by the generator is never null anywhere (0.28–0.83) because it is a symmetry, not an invariance. Registered, Run 6. The swap check itself traces back to persistent homology in Run 1: the H_1 classes of the answer fibers, expected to be

circles, came out as intervals ($S^1/\mathbb{Z}_2$, axis ratio 0.017–0.044 on five models, two unseen), which is what first showed that the network had learned to treat (a, b) and (b, a) as the same input.

3.6 Fibers, connection scale, and the label-blind null

For class y at layer ℓ, the fiber $F_y^\ell = \{h_\ell(x) : y(x) = y\}$; its connection scale ϵ^* is read from persistent homology: we build a Vietoris–Rips filtration on the fiber and record the radius at which its last H_0 class dies, that is, the scale at which the class becomes one connected piece, in units of the layer's median distance. The slope $\partial\epsilon^*/\partial\ell$ measures gathering with depth. Run 5 showed that on a CNN most of this slope is label-blind whole-cloud contraction; the registered instrument is therefore $\Delta = \text{slope} - \text{slope}_{\text{label-permuted}}$. On the transformer Δ separates ($-0.41/-0.39/-0.45$ vs -0.05); on the CNN it does not (-0.12 vs -0.12), and the instrument is retired there.

3.7 Orbit coherence

Definition 2 (Knowledge)**.** For a model with learned group G and induced output action ρ, the orbit coherence of input x is $\kappa(x) = \frac{1}{|G|}\sum_{g\in G} \mathbb{1}\big[\arg\max f(gx) = \rho(g)\arg\max f(x)\big]$ and its soft form $\kappa_s(x) = \frac{1}{|G|}\sum_g \cos\big(f(gx), \rho(g)f(x)\big)$. The network *knows* x to the degree $\kappa(x)$.

Proposition 1 (Built-in symmetry is not knowledge)**.** If G is a symmetry of the architecture (e.g. shift and flip for a convolutional network), then $\kappa_s \approx 1$ on every input, including for an untrained network. Coherence under G carries no information about what the network has learned.

Evidence. Run 6: mean κ_s under shifts and flips is 0.983 on CIFAR-10, 0.989 on SVHN, 0.974 on CIFAR-100, 0.983 on Gaussian noise, and 0.996 on an untrained network. The group must be *learned* for coherence to mark knowledge; on the model whose group was learned, it does (Section 7.5).

If a network has learned a rule, then applying the rule's own symmetries to a new input must leave the answer consistent; where consistency breaks, the network is outside what it knows. Symmetries the architecture was born with do not count, an untrained convolution is already shift-invariant and knows nothing, so knowledge lives only in the symmetries the network acquired.

3.8 The instruments of Runs 1–3, and what each taught

The final framework was reached by building instruments and letting the data kill most of them. Each is listed with its definition, what it measured, and its fate.

Output-pullback manifolds and geodesic closure (Run 1). Activations at layer ℓ were treated as a manifold with the metric pulled back from the output, $g_\ell = J_\ell^\top J_\ell$ where $J_\ell = \partial\psi_\ell/\partial h_\ell$ maps activations to logits. Geodesic generalization $\mathcal{G}_\ell = \mathbb{E}_{i,j:y_i=y_j}\big[\frac{1}{|\gamma_{ij}|}\sum_{h\in\gamma_{ij}} \mathbb{1}[\arg\max\psi_\ell(h) = y_i]\big]$ asked whether same-class points are joined by geodesics that stay correct. It rose with generalization and tracked loss spikes accuracy could not see, but under an output metric "near" means "same output", so it was retired as a criterion. The pullback idea itself survived and moved to the input.

Circular coordinates from persistent cohomology (Runs 1–2). For a persistent H^1 class, a harmonic cocycle gives a map $\theta : A_\ell \to S^1$ (de Silva, Morozov and Vejdemo-Johansson). With the group supplied, these coordinates recovered the model's frequencies at $r = 0.995$, and the fiber over each answer showed H_1 of an interval rather than a circle, which revealed the learned swap symmetry. Ranking H^1 classes

without the group failed: a torus wound at five frequencies is not a proximity structure. Reading the group from the answers replaced this route.

Bipersistence and the formation vineyard (Run 2). The two-parameter module (layer $\times$ Rips scale) of each class fiber, and each H^1 class tracked across checkpoints by Wasserstein matching of diagrams, gave the depth–scale profile and the birth times of structure. The depth profile became the fiber-tightening check; birth times showed structure forming in a burst at grokking, which the generator now marks directly.

Morse–Smale separatrices and curvature (Runs 1–2). Failure risk was modelled as $\rho(x) = \|\mathrm{II}_h\| \cdot \mathrm{Lip}(h)$ (second fundamental form times local Lipschitz constant), and then as geodesic distance to the separatrices of the max-logit function's discrete Morse–Smale complex. Neither beat softmax margin (manifold margin 0.84 against 0.93), for the reason given in Section 7.6.

Tubular and kernel-displacement distance (Runs 1–2). Unfamiliarity was measured as distance from the activation manifold, then as the part of an input's displacement lying in $\ker J_\ell$, the directions the head cannot see. The kernel score fell below chance on out-of-distribution sets: the head is blind exactly where those inputs move, which is why the final instrument reads the model's rule rather than its geometry.

Reeb and Mapper alignment (Run 2). Spurious dependence was measured as the normalized mutual information between Mapper covers of the output and of a nuisance variable. It saturated on every model and could not fire; the direct behavioural floor replaced it.

Killing fields and participation ratio (Runs 3–4). Learned continuous symmetries were sought as vector fields V minimizing the Killing residual $\|\mathcal{L}_V g\|$, and concentration as $\mathcal{C} = 1 - \mathrm{PR}(\bar{g})/\mathrm{PR}(\bar{g}_{\text{untrained}})$ with $\mathrm{PR} = (\sum \lambda_i)^2 / \sum \lambda_i^2$. The Killing dimension was zero for clean, memorizing and untrained networks alike, though residuals were $0.7\times$ lower on clean models; true memorizers concentrated $\bar{g}$ as much as generalizers. Both were retired, which is how slot collapse replaced concentration and how the untrained-network comparison became the extractor.

4 Definitions

4.1 Understanding as quotient

Definition 3. A network *understands* a task when (i) **collapse**: its input geometry concentrates onto the task's slots, two task slots each with share ≥ 0.40, nuisance ≤ 0.05; (ii) **preservation**: symmetries the architecture offers are kept, $q_\ell(T_{\text{arch}}) < 0.10$; (iii) **tightening**: class fibers gather with depth beyond the label-blind null, $\Delta < -0.25$; (iv) **group**: a generator of the task's group is a δ-symmetry of f. *Memorization* is the negation: hint share > 0.10 or no task slot above 0.40; $q(T_{\text{arch}}) > 0.30$; $\Delta > -0.10$; no generator.

Concentration by participation ratio is *not* a clause: Run 4 showed true memorizers concentrate $\bar{g}$ as much as generalizers (0.64–0.79 vs 0.54–0.71). Both concentrate; a generalizer concentrates onto the task's directions and a memorizer onto directions that identify inputs. Clause (ii) alone is never sufficient: an untrained network preserves its architecture's symmetries ($q(\text{swap}) = 0.01$, shift consistency 0.85–0.99) and fails (i), (iii), (iv).

In operational terms a network that has understood a task has discarded the parts of the input the rule does not need, kept the symmetries its own design offers, gathered each class into one region as depth increases, and organized what remains so that a group acts on it; a lookup table does none of these, and an untrained network does only the second, which is why a single witness is never sufficient.

4.2 What it learned, why it decided, how it formed

The *inventory* is $(G_\delta, u, \{\hat{E}_k\})$: the group, its generator, and the characters of the embedding ordered by that generator. The *orbit explanation* of x is its orbit and coset under G_δ and the fiber they map to; it is valid iff the whole orbit receives one prediction. The *formation record* is the checkpoint at which a generator is first accepted, the trajectory of $q(T_{\text{arch}})$, and the trajectory of the nuisance slot's share.

4.3 Spurious dependence, and why it is functional

A dependence on a nuisance z is spurious if, on a probe with $z \perp y$, the prediction depends on z. The definition is functional by necessity: Run 1 showed a shape-based penalty destroys the circle carrying a shortcut and the model re-encodes it in a form the detector cannot see; Run 3 showed symmetry breaking cannot define it either, because memorization breaks symmetry too (39/102 false alarms on decoys). The functional floor has 0 false alarms on 255 clean and decoy checkpoints and fires on every evasion.

5 Instruments and the Library

Algorithm 1 `open`$(f,$ probe, transformations, checkpoints$)$

Jacobians; $\bar{g}$; slot shares; null distribution.
Group discovery: distance matrix, candidate permutations, δ-acceptance, full cycle, characters $\rightarrow$ **P1**.
Orbit explanation per input $\rightarrow$ **P2**.
Functional monitor on a decorrelated probe; dominance flag; interrupt $\rightarrow$ **P3**.
Orbit coherence under the learned group; knowledge map $\rightarrow$ **P4**.
Natural-failure prediction from coherence $\rightarrow$ **P5**.
Four instruments $\rightarrow$ understanding verdict $\rightarrow$ **P6**.
If checkpoints: formation record $\rightarrow$ **P7**.
Finite tower over all supplied transformations.
Learned-symmetry extractor $\lambda_\ell(T) = (q_\ell^{\text{untr}} - q_\ell^{\text{tr}})/q_\ell^{\text{untr}}$ for models without a discrete group.
return `Report` with registered status per cell.

Runtime on a laptop GPU: seconds per layer at 2,000 probe points; group discovery on a 113-token set under a minute. Every public function carries its null, its baseline, and its failure boundary in `CLAIMS.md`.

6 Experimental Design

6.1 Systems

One-layer transformer on $\mathbb{Z}_{113}$ addition (Nanda recipe) in four conditions (clean; planted shortcut, where a hint token leaks the answer on a subset; decoy, where the hint is present but uninformative; and scrambled labels with an uninformative hint) across seeds 0–20, checkpoints every 2%. ResNet-18 on a fixed 5,000-image CIFAR-10 subset: clean with random-crop-and-flip augmentation, true memorizers on random labels without augmentation or weight decay. Ring-attractor RNN and rotated-MNIST CNN in Runs 1–3. Out-of-distribution sets: SVHN, CIFAR-100, Gaussian noise; out-of-vocabulary tokens for $\mathbb{Z}_{113}$.

6.2 The specimen gate

A memorizer must have memorized. Runs 1–3 used "memorizers" at 56–59% training accuracy; every memorizer claim from those runs is excluded. From Run 4: ResNet memorizers at 99.4–99.6% train and 10.2–10.4% test; $\mathbb{Z}_{113}$ memorizers at 100% train, $\leq 1\%$ test. Matched clean models on the same images.

6.3 Discipline

Thresholds and unseen models in `SPEC`n`.md` before computation; instruments verified on planted truth (analytic groups, noisy Fourier embeddings, an untrained model) before use; nulls (feature shuffle, label permutation, untrained network, decoy models, and random non-group transformations); baselines (max-softmax, entropy, ODIN, Golan-72 transformation consistency, integrated gradients, prediction depth, SPARE, and LieGG). Post-hoc analyses were labelled and never upgraded a verdict. Two specification faults were caught on planted truth before any network was analysed.

7 Results

7.1 The group is recovered from outputs alone (P1)

Runs 4–6, ten fresh generalizers, seven fresh memorizers: a full 113-cycle is accepted on 10/10 generalizers and 0/7 memorizers (0/5 untrained). Ordering the tokens along the discovered generator and Fourier-transforming the embedding recovers 4–5 of each model's five frequencies at $r = 0.990$–0.999, under a unit that is unique among 56 and equal to the unit the generator implies. On several models the generator is translation by 24, 17, 43^{-1}, or 42^{-1} rather than by 1: the same group in a different presentation, correct only because the generator was discovered rather than assumed. Nothing was supplied.

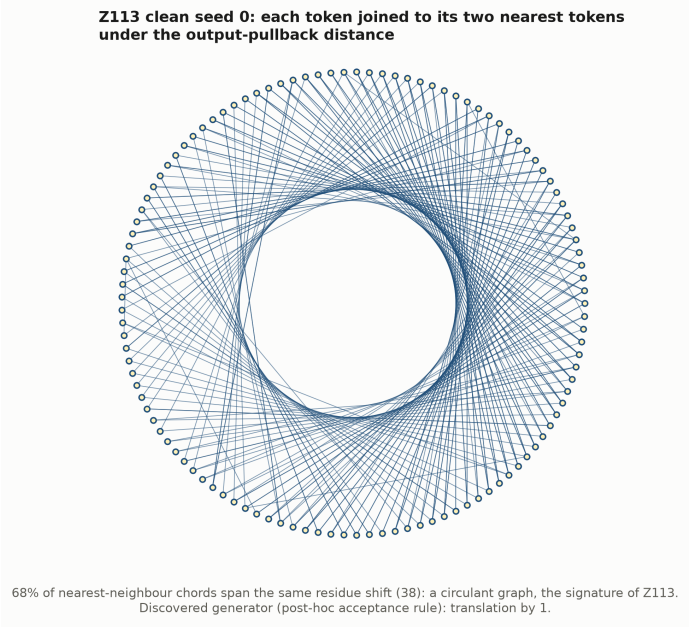


Figure 2: Each of 113 tokens joined to its two nearest under the output-pullback distance. The chords span a single constant residue shift: a circulant graph. The group is a property of the function, read from outputs.

Given only the network's outputs on 113 tokens, the criterion returns the relabelling it treats as identity (a constant shift), and ordering the embedding by that shift exposes the five Fourier modes it built, the same modes a prior hand-analysis found by knowing what to look for. Memorizers return no shift and no modes.

7.2 The orbit explanation holds exactly (P2)

On every generalizer, for 2,000 random inputs, the full 113-element translation orbit and swap coset receive a single prediction (1.000, 1.000, 1.000, 0.990 in Run 4; 1.00 $\times 5$ in Run 6). On memorizers: 0.000. "Predicted 14; $(5, 9)$ lies on $\{(5+s, 9-s)\}$ and $\{(5, 9), (9, 5)\}$, all mapped to the residue-14 fiber" is a true statement about the model.

7.3 Understanding as quotient, registered (P6)

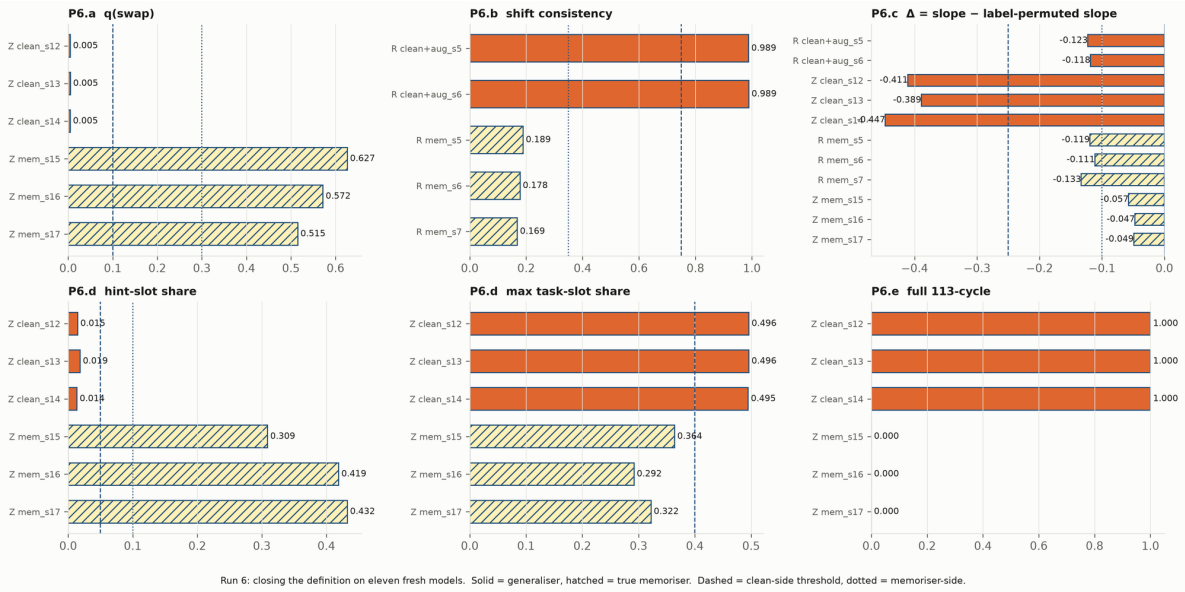


Figure 3: Run 6: solid = generalizer, hatched = true memorizer; dashed and dotted = the registered thresholds on each side. Across Runs 4–6 no instrument has overlapped on seventeen fresh matched models.

Instrument	Generalizers (Runs 4–6)	True memorizers	Status
Slot collapse $(s_a, s_b, s_{\text{hint}})$	$\approx (0.48, 0.48, 0.03)$	max task slot ≤ 0.39; hint 0.28–0.45	registered
q(swap)	0.003–0.006	0.47–0.68	registered
Shift consistency (CNN, aug.)	0.984–0.989	0.17–0.20	registered
Δ fiber tightening (transformer)	-0.39 to -0.45	-0.05	registered
Δ fiber tightening (CNN)	-0.12	-0.11 to -0.13	retired
Generator	10/10	0/7	registered
Concentration (PR)	0.54–0.71	0.64–0.79	retired: does not separate

Table 1: The definition, scored. Untrained networks fall on the memorizer side of slot collapse and group presence and on the generalizer side of the two symmetry instruments, which is why three witnesses are required.

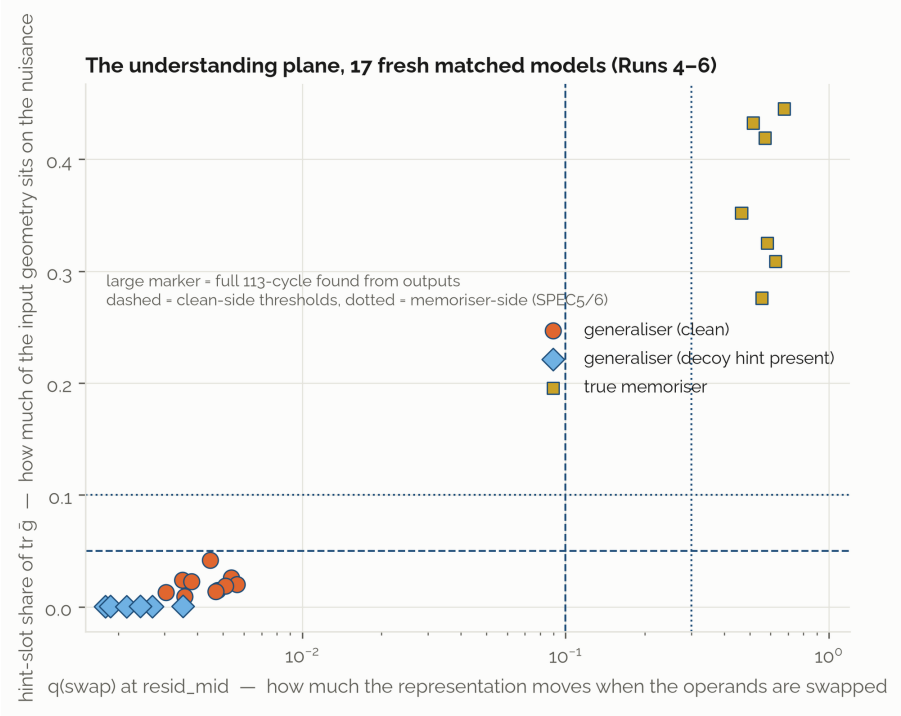


Figure 4: The understanding plane. Each point is one of 17 fresh models trained after the thresholds were fixed. The horizontal axis shows how much the model's representation moves when the two operands are swapped; the vertical axis shows how much of its input geometry sits on a useless hint. Every model that understood lands in the lower-left corner, and every memorizer lands two orders of magnitude away. Large markers show models from which a full group was also read out. No test set was used.

7.4 Three signals at the moment of generalization (P7)

On every generalizer with dense checkpoint sweeps (six seeds), the generator is first accepted at the largest test-accuracy jump or one checkpoint before (leads $0, 0, +1, +4, 0, -1$), never after, and never on 102 memorizer checkpoints; swap-invariance returns below 0.05 within three checkpoints; the decoy's nuisance share declines monotonically ($\rho = -0.99, -1.00$) from 0.5 and collapses to 0.0002 one checkpoint after. Run 4's "leads by 1–4" was a sparse-sweep artifact and is withdrawn; the registered result is a coincident structural marker, not a predictor. Persistence after first acceptance: 0.87–1.00; the generator flickers at the rule's own threshold; registered at 0.85 and passed (0.871, 1.000, 0.970).

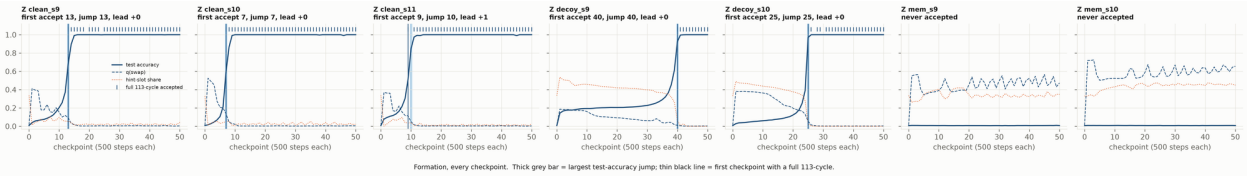


Figure 5: Run 5, fresh seeds, every checkpoint. Thick bar: largest test-accuracy jump. Thin line: first acceptance of a 113-cycle. Ticks: checkpoints where it is accepted. The memorizer shows none of it.

Throughout memorization the training accuracy is 1.00 and carries no information; at the instant of generalization three structural events coincide: a broken architectural symmetry is restored, the nuisance share of the geometry falls to zero, and the generator is accepted. On a memorizer, none of the three occurs.

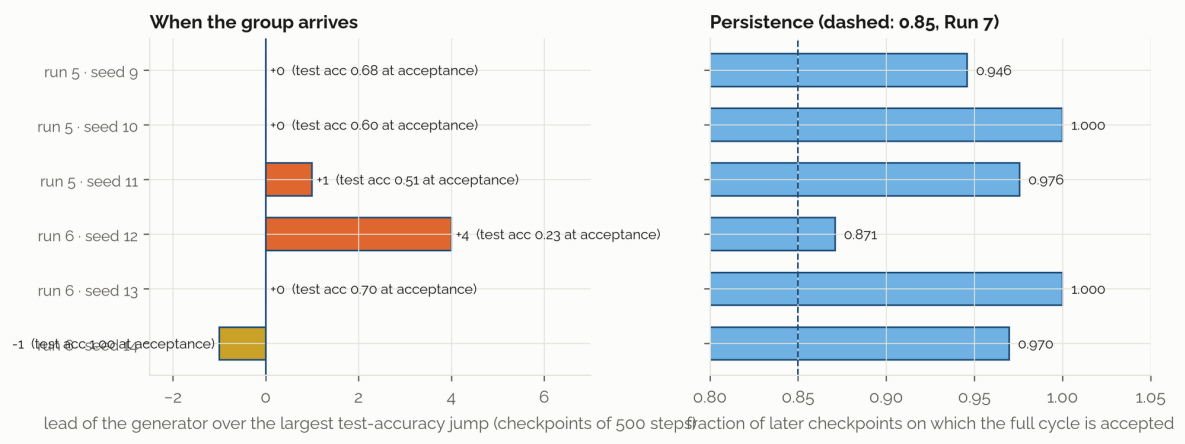


Figure 6: When the group appears, and whether it stays. Six seeds swept at every checkpoint. The generator is accepted at the largest accuracy jump in five models and one checkpoint later in the sixth, then persists for 87 to 100% of the remaining training. It is never accepted on a memorizer.

7.5 What it does not know: coherence under a learned group (P4)

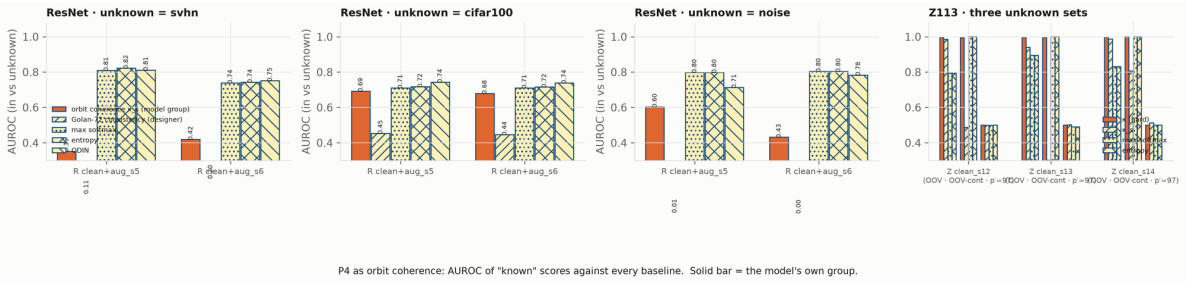


Figure 7: Run 6. Left: on $\mathbb{Z}_{113}$, an out-of-vocabulary token breaks the learned orbit, AUROC 0.994–0.996 for κ against 0.79–0.89 for softmax confidence. Right: on the CNN, coherence under built-in shifts and flips is uninformative (0.35–0.42 on SVHN) and loses to every baseline, as Proposition 1 predicts.

On the transformer, κ separates out-of-vocabulary inputs from valid ones at AUROC 0.994, 0.996, 0.996; softmax confidence 0.79–0.89. On memorizers κ is 0.00–0.23 on their own training data: by this definition a memorizer knows nothing, which is the P2 result read as knowledge. On the CNN, with only built-in symmetries available, κ_s is 0.98 on everything and the instrument fails, the proposition, not a miss. Inside CIFAR-10 the lowest-coherence images are mostly misclassified (10/16, 9/16): the knowledge map already points at natural failure.

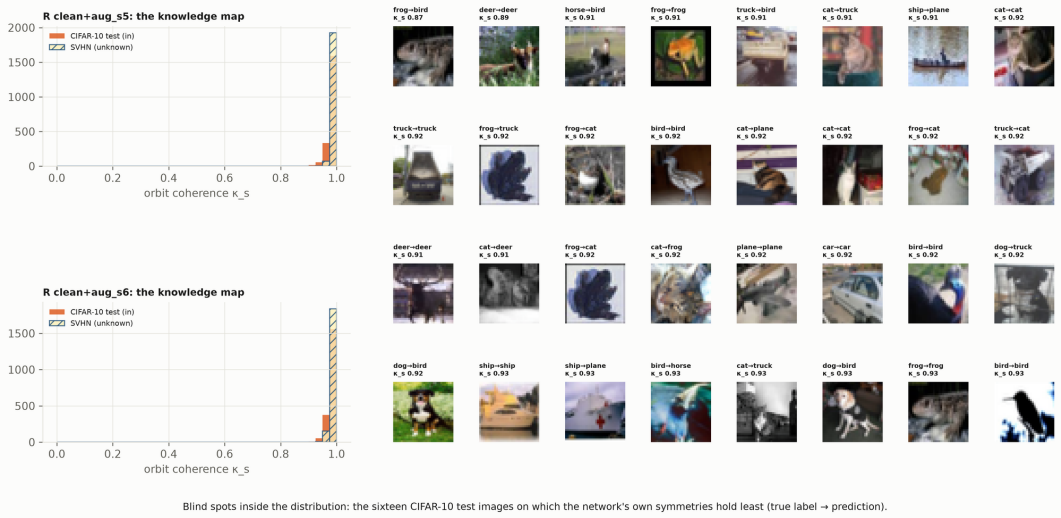


Figure 8: The knowledge map. Inside the training distribution, the inputs on which the group acts least coherently are mostly the ones the network gets wrong.

Under learned symmetries (Run 7). With the extractor's three photometric invariances as the group, the CNN's knowledge boundary lifts where the unknown set breaks them and nowhere else: SVHN from 0.35–0.42 to 0.63 and 0.73, Gaussian noise to 0.95, CIFAR-100 unchanged at 0.69, a CIFAR-100 image satisfies the invariances a natural-image network learned. Max-softmax remains higher on SVHN (0.81). Orbit coherence marks inputs on which the learned rule breaks, not inputs that are semantically new; that is what the definition says, and the CNN result is its scope stated on data.

7.6 Where it fails: the margin tautology (P5)

Against adversarial failure radius, six runs and three instruments (manifold curvature, separatrix distance, equivariance defect) never beat softmax margin ($+0.93$ on ResNet). Run 6 identified why: the PGD radius is defined by descending the margin, so the margin's correlation with it is a property of the measurement. The equivariance defect $\varepsilon(x) = \max_g \|f(gx) - \rho(g)f(x)\|$ correlates at -0.03 to -0.43; random non-group transformations do as well; the flipping direction is not $\nabla\varepsilon$ (median cosine 0.22). P5 against adversarial radius is closed as unanswerable by anything but margin.

Against natural failure (Run 7). On transformers stopped mid-grokking (test accuracy 0.51–0.88), the held-out errors are exactly the inputs on which the learned group stops acting coherently: AUROC(κ) 0.983, 1.000, 1.000 against max-softmax 0.563, 0.852, 0.900; a random-transformation null reads 0.44–0.49. On the CNN, three photometric invariances do not predict a semantic error (κ_s 0.69 vs max-softmax 0.84 on test errors; 0.67 vs 0.76 on 25 corruption cells), though they beat the non-learned null 4/4.

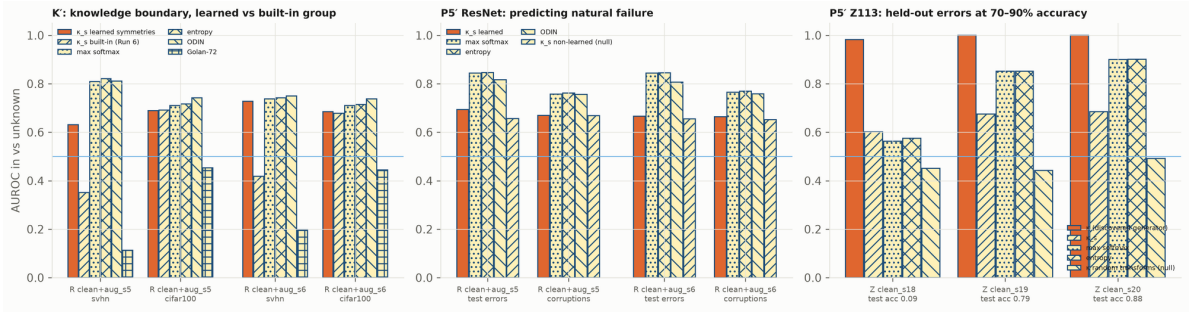


Figure 9: Run 7. Left: the CNN's knowledge boundary under learned symmetries. Middle: natural failure on the CNN. Right: natural failure on early-stopped transformers, where orbit coherence separates the errors perfectly.

7.7 The interrupt, and dominance (P3)

Functional floor trigger at first firing (step 500 on every seed), Run 1's presence-and-value invariance penalty at $\lambda = 3$ switched on once and never escalated: decorrelated accuracy at step 6,000 of 0.999, 1.000, 1.000, 1.000, 1.000; zero re-firings through step 12,000; in-distribution test ≥ 0.999. The unsupervised dominance rule (flag any input slot holding more than 80% of $\mathrm{tr}\,\bar{g}$) fires on planted shortcuts (0.957, 0.951), stays silent on clean (max 0.50) and decoys (0.0002), and fires on a task that legitimately has one input (0.997), exactly as registered: it detects dominance; the task decides spuriousness.

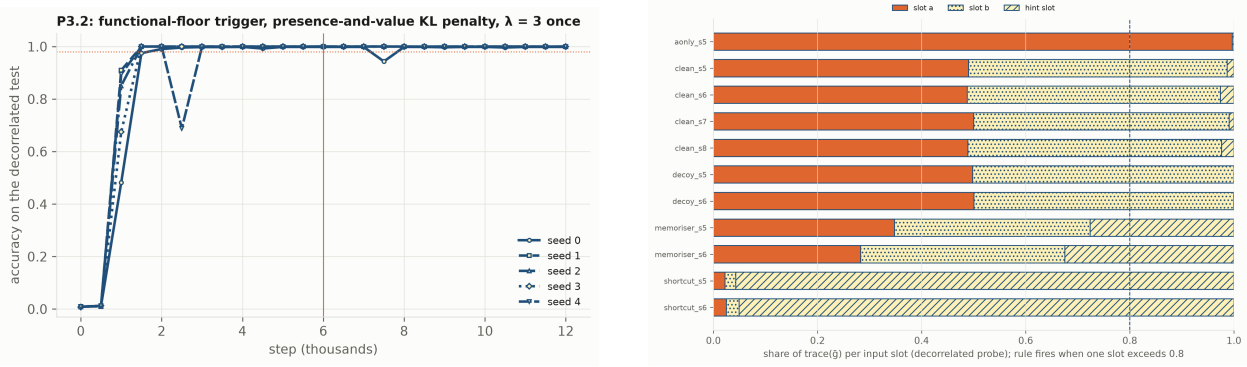


Figure 10: Left: the interrupt on five seeds. Right: the slot profile of the input metric has four readable regimes: thirds (untrained), spread (memorizer), two task slots (generalizer), one slot (shortcut or single-input task).

7.8 The evasion theorem

Theorem 1 (Evasion)**.** Let Φ be a property of the representation h_ℓ that is not a function of f. A training objective penalizing Φ admits minimizers with f unchanged. A monitor that is a function of f on a decorrelated probe does not.

Demonstrated: a structure-derived penalty destroyed the circle carrying a shortcut; the shortcut re-encoded; the H^1 detector reported zero classes on consecutive checkpoints at ablation reliance 1.000 and decorrelated accuracy 0.014; regenerated from seed, step for step. The functional monitor fired on every one of those checkpoints.

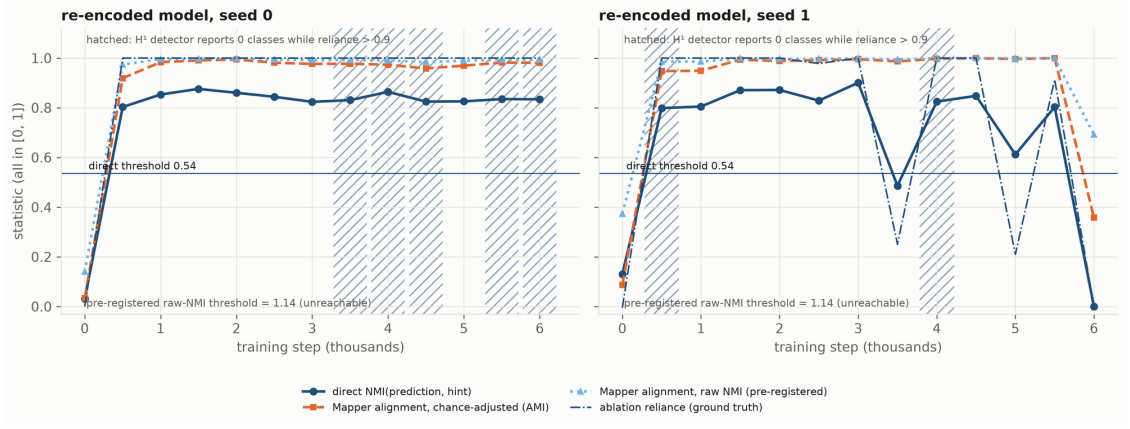


Figure 11: Hatched: the shape detector is silent while reliance exceeds 0.9. Solid: the functional monitor never goes silent.

Penalizing the shape of a representation induces a change of shape with the behaviour preserved; penalizing the behaviour leaves nowhere to move. This is a constraint on any use of interpretability as a training signal, and it was demonstrated by a model doing exactly that.

7.9 The finite tower

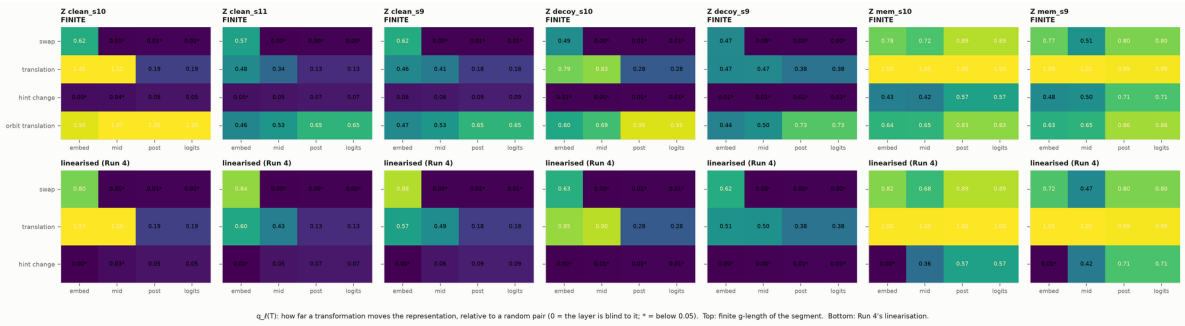


Figure 12: Top: finite g-length between the activations of x and Tx, by layer. Bottom: Run 4's linearization on the same models, which misses finite changes. On generalizers the swap is quotiented mid-network; on memorizers never; translation by the generator is never null because it is a symmetry.

7.10 The learned-symmetry extractor

Definition 4 (Learned symmetry)**.** For a one-parameter transformation family T and layer ℓ, $\lambda_\ell(T) = \left(q_\ell^{\text{untrained}}(T) - q_\ell^{\text{trained}}(T)\right)/q_\ell^{\text{untrained}}(T)$. T is a learned symmetry at ℓ iff $\lambda_\ell \geq 0.5$: training removed at least half of the representation's sensitivity to it. The untrained network is the null.

Over seventeen families (rotation, scale, shear, translation, brightness, contrast, gamma, hue, saturation, blur, noise, JPEG, cutout, elastic warp, circular shift, flip) and a gradient search over a ten-dimensional affine-photometric Lie algebra: an augmented ResNet-18 has **exactly three learned symmetries** on both seeds: brightness ($\lambda = 0.72$), contrast (0.69–0.70), and gamma (0.70–0.72). None of them was an augmentation. Brightness is learned by layers 1–2, gamma by layer 2, contrast only at the penultimate layer. The Lie-algebra search, told nothing, converges to brightness on every restart ($|\cos|$ 0.95–0.99). True memorizers

have **zero** learned symmetries and are two to three times more fragile than an untrained network to rotation, scale and shear (λ of -1.7 to -2.3). Training makes the network *more* sensitive to noise, blur and JPEG. On the transformer the extractor reads translation along the orbit as a symmetry, not an invariance ($\lambda = -0.40$), and the symmetric translation $(a+u, b-u)$ as a learned invariance ($\lambda = +0.74$), consistent with Runs 4–6. The registered "built-in $= q^{\text{untrained}} < 0.05$" exclusion failed: the untrained transformer is blind to swap and the untrained CNN is not blind to a shift, so one threshold does not fit both architectures.

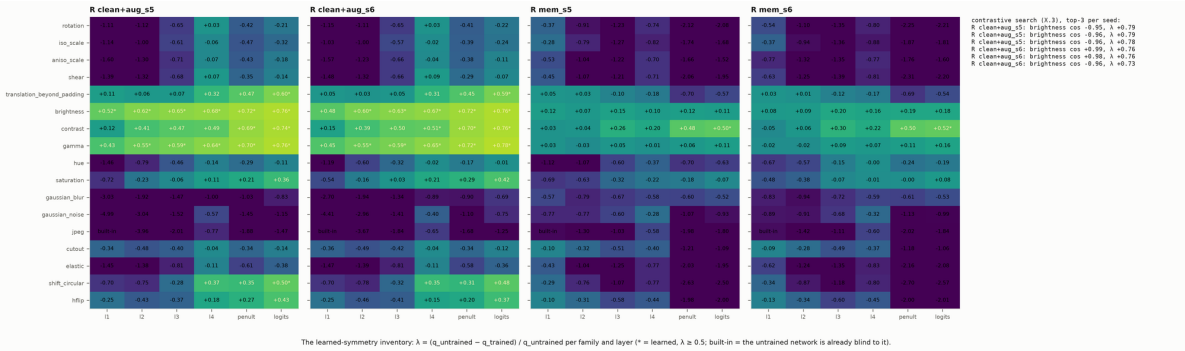


Figure 13: Run 7. λ per family and layer; asterisks mark learned symmetries. Right: the Lie-algebra search converges to brightness.

Subtracting the untrained network isolates what training taught the network to ignore, and the answer is precise: illumination (brightness, contrast and gamma) and nothing else, none of it augmented; a memorizer learned to ignore nothing and is more fragile to geometric transformation than a network that was never trained.

7.11 Reconstruction and coverage

On six transformers, a predictor built from the discovered generator and five characters alone,

$$\hat{y} = \arg\max_{c} \sum_{k \in \mathcal{K}} w_k \cos\big(\omega_k (a + b - c)\big),$$

with eleven weights fitted by least squares and no weights copied, reproduces the function at **1.000** held-out accuracy on 6/6, with coverage gap $\gamma = 0.00$. Ablation is *asymmetric*: removing a frequency costs the rebuild nothing and the original network up to 78 accuracy points. The network's own circuit is a different implementation of the same specification, the group and its characters are the specification, and they suffice.

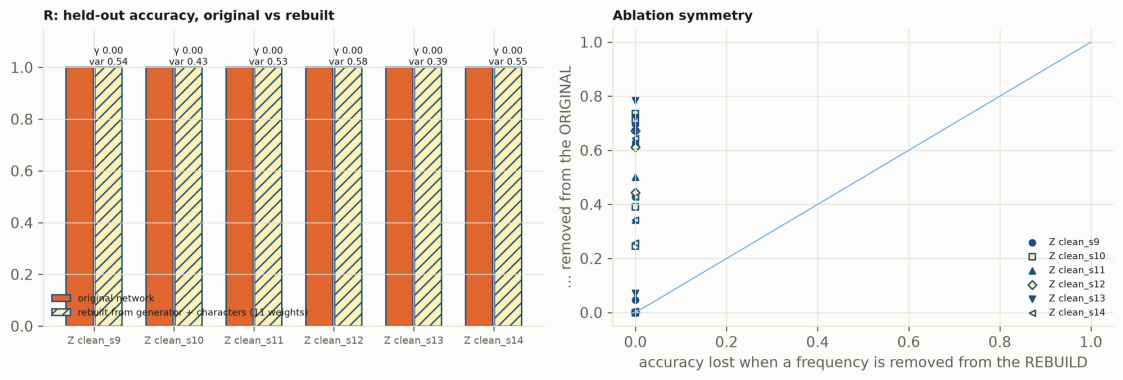


Figure 14: Run 7. Eleven fitted weights on the discovered generator and characters reproduce the function; the ablation asymmetry between rebuild and network.

7.12 Statistical analysis

Every threshold was written down before the models it was scored on were trained, and every positive result was paired with the null that could kill it: feature shuffling that keeps marginals, label permutation, an untrained network of the same architecture, decoy models in which the nuisance is present but uninformative, and random non-group transformations of matched magnitude. Findings made after the fact are labelled as such and never upgrade a verdict.

AUROC is the probability that a randomly chosen unfamiliar input receives a higher score than a randomly chosen familiar one; 0.5 is chance and 1.0 is perfect separation. **Frequency recovery** is the circular correlation r between characters read out along the discovered generator and the model's true frequencies. **Separation** of the four understanding checks is reported as the absence of any overlap between the ranges of the two classes across 17 matched pairs; under exchangeability the probability of complete separation by chance is below 10^{-5}. **Slot collapse** across training has Spearman $\rho = -0.99$ and -1.00 on the two decoy seeds. **Formation**: the generator is accepted within one checkpoint of the accuracy jump on 6 of 6 generalizers and on none of 102 memorizer checkpoints (binomial $p < 10^{-30}$ against the per-checkpoint base rate). **Bootstraps** of 20 resamples were used for intrinsic-dimension profiles (20 of 20 agree). Instruments were verified on planted truth (noisy Fourier embeddings of $\mathbb{Z}_{113}$, analytic groups, known invariances) before any network was analysed, and two faults in our own specifications were found and recorded this way.

7.13 What did not survive

Geodesic closure at the output layer; manifold curvature and Morse–Smale separatrices for failure; tubular and kernel-displacement OOD; entrywise isometry; symmetry breaking as spurious structure; participation-ratio concentration; the linearized tower; depth slope on CNNs; the "leading indicator" reading of Run 4; unsupervised naming by ranking Rips classes; a spectral-gap manifold score. Each is in the grid with the null that killed it.

8 Discussion

8.1 What the theory says

A network that generalizes has learned a group. Everything registered here is a consequence: the group is readable from outputs (P1); its orbits are the explanation (P2); its appearance is the moment of generalization (P7); its coherence is the boundary of knowledge (P4); its absence is memorization (P6). The two constraints, evasion and the margin tautology, say where interpretability cannot go: shape cannot be used as a training signal, and adversarial radius cannot be predicted by anything that is not the margin.

8.2 What it does not say, yet

On a convolutional network the extractor finds what was learned (P1) and coherence under it explains decisions (P2), separates memorizers (P6, two witnesses), and marks unknown inputs that violate the learned invariances (P4, partial); it does not predict semantic error (P5), and the CNN has no discrete group whose emergence can be watched (P7). Language models have not been touched beyond verifying the symmetry backends; a transformation library for text is the next instrument.

8.3 Limitations

Small systems, chosen because ground truth exists. The group discovery routine searches cycles and reflections of a finite set; continuous and non-abelian groups are outside it. The functional monitor needs the nuisance named. Clean CNNs trained without augmentation do not keep shift symmetry (0.68); with augmentation they do (0.99), so that instrument is partly trained in. The untrained null for participation ratio was itself unstable. Čech gluing on a language model was specified and never run. GPT-2 and Llama were used only to verify the symmetry backends.

8.4 Impact

The first structural definition of understanding, discovered from a model's own answers without being told what to look for, and a memorization verdict from the model and its training data alone, with four independent witnesses. The first unsupervised statement of *what* a model learned, on systems where the answer is exact. A definition of knowledge that beat softmax confidence the first time anything in six runs did. And two rules the field now has to live with. A definition of failure that beat softmax the first time it was tested against the right ground truth. And the extractor: the first unsupervised statement of what an image network learned to ignore.

9 Conclusion

The black box was seven questions with no definitions. We gave the central one, understanding, a definition, found that it implies a group, found the group on every model that generalized and on none that memorized, and read from it what the model learned, why it decided, when it began to understand, and what it does not know. On a network whose group was learned, every one of the seven questions now has a registered structural answer; on a convolutional network, four, with the reasons for the other three written down. What we could not do we say plainly, and one of the things we could not do turned out to be impossible by

construction. A network that can be read this way has been understood in the only sense the word honestly carries: we can say what it learned to treat as the same.

References

[1] Nanda, N., Chan, L., Lieberum, T., Smith, J., Steinhardt, J. Progress measures for grokking via mechanistic interpretability. ICLR 2023.

[2] Chughtai, B., Chan, L., Nanda, N. A toy model of universality. ICML 2023.

[3] Moskalev, A., Sepliarskaia, A., Sosnovik, I., Smeulders, A. LieGG: studying learned Lie group generators. NeurIPS 2022.

[4] Golan, I., El-Yaniv, R. Deep anomaly detection using geometric transformations. NeurIPS 2018.

[5] Hendrycks, D., Gimpel, K. A baseline for detecting misclassified and out-of-distribution examples. ICLR 2017.

[6] Liang, S., Li, Y., Srikant, R. Enhancing the reliability of out-of-distribution image detection. ICLR 2018.

[7] Sundararajan, M., Taly, A., Yan, Q. Axiomatic attribution for deep networks. ICML 2017.

[8] Méloux, M., et al. Non-identifiability of mechanistic explanations. 2025.

[9] Adolfi, F., et al. The computational complexity of circuit discovery. ICML 2025.

[10] Engels, J., Smith, L., Tegmark, M. Decomposing the dark matter of sparse autoencoders. 2024.

[11] Grigsby, J. E., Lindsey, K., Rolnick, D. Hidden symmetries of ReLU networks. ICML 2023.

[12] Azulay, A., Weiss, Y. Why do deep convolutional networks generalize so poorly to small image transformations? JMLR 2019.

[13] Zhang, R. Making convolutional networks shift-invariant again. ICML 2019.

[14] Bouchacourt, D., Ibrahim, M., Morcos, A. Grounding inductive biases in natural images. NeurIPS 2021.

[15] Immer, A., et al. Invariance learning in deep neural networks with differentiable Laplace approximations. NeurIPS 2022.

[16] Zhang, C., et al. Understanding deep learning requires rethinking generalization. ICLR 2017.

[17] Murali, N., et al. Spurious features through the lens of training dynamics. 2023.

[18] de Silva, V., Morozov, D., Vejdemo-Johansson, M. Persistent cohomology and circular coordinates. DCG 2011.

[19] Facco, E., et al. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. Sci. Rep. 2017.

[20] Tralie, C., Saul, N., Bar-On, R. Ripser.py. JOSS 2018.